

Non-Desmos - Problem Solving & Data Analysis

10 original questions | Calculator permitted

Complete the questions before turning to the answer explanations. Mark any guess, even if it turns out to be correct. For math, show your setup and check the requested quantity. For reading, identify the exact supporting detail; for grammar, identify the clause or subject.

These questions were authored for this workbook from the lesson topics. They are not official SAT questions or copies of the course's paid practice.

Problem Solving & Data Analysis | 10 questions

[Lesson video: Problem Solving & Data Analysis](#)

1. The table shows the lengths of 10 seedlings. What is their mean length, in centimeters?

Length (cm)	Frequency
10	2
14	5
18	3

Answer: _____

2. Data set A is 4, 6, 8, 10, 12. Data set B is formed by adding 30 to every value in A. Which statement is true?

- A) The mean and standard deviation both increase by 30.
- B) The mean increases by 30, and the standard deviation is unchanged.
- C) The mean is unchanged, and the standard deviation increases by 30.
- D) The mean and standard deviation are unchanged.

3. The values 2, 5, 7, 8, and 13 form a data set. The value 25 is added. By how much does the median increase?

Answer: _____

4. The table gives temperatures of two liquids. Between which pair of consecutive times is the magnitude of Liquid A's average cooling rate smaller than that of Liquid B?

Time (min)	A (°C)	B (°C)
0	80	80
10	60	70
20	48	62
30	44	56

- A) 0 to 10 minutes
- B) 10 to 20 minutes
- C) 20 to 30 minutes
- D) The magnitudes are equal in every interval.

5. A square panel has area 4,500 square centimeters. What is its area in square meters? (1 meter = 100 centimeters.)

Answer: _____

6. After a 20% discount and a 5% tax on the discounted price, a desk costs \$252. What was the original price, in dollars?

Answer: _____

7. A 12-liter solution contains 25% salt by volume. Pure water is added until the concentration is 15%. How many liters of water were added?

Answer: _____

8. One person is chosen at random from participants who completed a workshop. What is the probability that the person is in Group B?

Group	Completed	Did not complete
A	24	16
B	18	12

- A) $18/70$
- B) $18/30$
- C) $18/42$
- D) $30/70$

9. A random sample of 400 members of one city's public library system finds that 62% prefer digital notices. The margin of error is 4 percentage points. Which conclusion is supported?

- A) Exactly 62% of all residents prefer digital notices.
- B) A plausible range for the proportion of library-system members who prefer digital notices is 58% to 66%.
- C) At least 96% of library-system members prefer digital notices.
- D) The result applies to every library system in the country.

10. Researchers recruit 120 volunteers and randomly assign half to a new study method and half to an existing method. Both groups study the same material for the same amount of time. Which feature most directly supports a causal conclusion about the methods?

- A) The volunteers came from one school.
- B) The groups were randomly assigned.
- C) The study included 120 people rather than 12.
- D) The study reported an average score.

Answers & explanations

Count correct answers, but also review correct guesses. Redo missed questions after a day without looking at the solution.

1: 14.4 2: B 3: 0.5 4: C 5: 0.45 6: 300 7: 8 8: C 9: B 10: B

Why each answer works

1. Answer: 14.4. Use the frequencies as weights: $(10 \cdot 2 + 14 \cdot 5 + 18 \cdot 3)/10 = 144/10 = 14.4$. Averaging the three listed lengths without their frequencies would be incorrect.

2. Answer: B. Translation increases every center measure by 30 but preserves all distances from the mean. Therefore standard deviation and range are unchanged.

3. Answer: 0.5. The original median is 7. The new ordered set has six values, so its median is $(7 + 8)/2 = 7.5$. Adding a high outlier can change the median; calculate rather than assume.

4. Answer: C. Over the first two intervals A drops 20 and 12 degrees, whereas B drops 10 and 8. Over the last interval A drops 4 and B drops 6, so A's rate magnitude is smaller then. A lower temperature alone would not determine its cooling rate.

5. Answer: 0.45. Square the linear conversion: $1 \text{ m}^2 = 10,000 \text{ cm}^2$. Therefore $4,500/10,000 = 0.45 \text{ m}^2$. Dividing by 100 would convert a length, not an area.

6. Answer: 300. Let P be the original price. $P(0.80)(1.05) = 252$, so $P = 252/0.84 = 300$. The percentage factors multiply; they do not simply cancel.

7. Answer: 8. Salt remains $12(0.25) = 3$ liters. If w liters of water are added, $3/(12 + w) = 0.15$. Hence the total is 20 liters and $w = 8$.

8. Answer: C. Restrict the population to completers: $24 + 18 = 42$ people. Of those, 18 are in B. The total group size and entire participant count are not the correct denominator.

9. Answer: B. The interval is $62\% \pm 4$ percentage points. Random sampling supports inference to the sampling population, library-system members, not all residents or the whole country.

10. Answer: B. Random assignment helps balance other influences between treatment groups and supports a causal comparison under a well-controlled design. It differs from random sampling, which supports population generalization.

For each missed or guessed question, record the skill, the condition you overlooked, and one specific action for next time.